

Hepatocyte Zonation in MASLD Progression

From weak indicators to the molecular event, the pericentral program turns OFF.

Healthy liver is spatially zoned — pericentral and periportal hepatocytes run opposite metabolic programs. We assigned every hepatocyte a pericentral↔periportal coordinate using two independent rulers (a curated gene-set and a label-free PCA axis), then transferred them to a 47-donor MASLD cohort (~69,000 hepatocytes). Every readout uses the DONOR as the unit of inference.

Methods note: the scored matrix is the SCT-corrected assay, so per-cell sequencing depth is already largely equalized (~4-5k counts). Confounder controls are parked for now (CONFOUNDERS_PLAN.md); this report is the biology.

The argument, in one page

The question

Healthy hepatocytes are zoned. Does that zonation break down in MASLD — and if so, HOW? Is it (a) the genes switching OFF (turn-off), or (b) the genes staying on but losing their spatial gradient (de-zonation)? These are different biology.

Why summary statistics did not settle it

The first instinct is to summarise each donor with one number — the spread of its cells along the zonation axis, the inter-quartile range, the pericentral-periportal anti-correlation. These are INDICATORS, and they are weak: the per-donor coordinate distribution is a broad single hump whose width changes only modestly and non-monotonically with disease. You cannot read the mechanism off them.

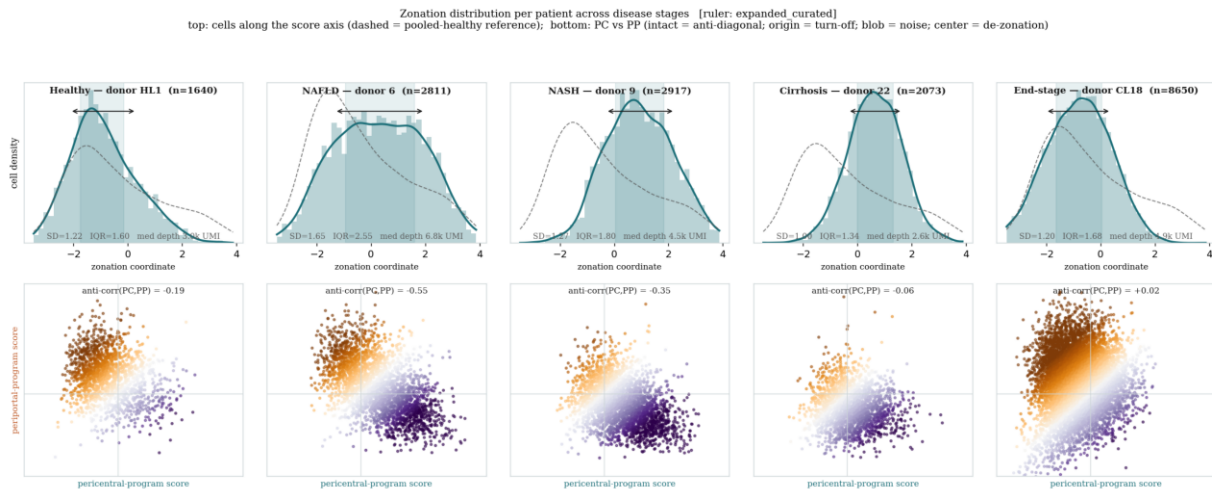
So we went into the biology, gene by gene

Instead of summarising, we looked at the actual expression — per zone, per gene, per stage and per fibrosis grade. The answer is clear and consistent across every view:

- The PERICENTRAL program TURNS OFF — its genes are silenced, expression level drops ~34-37% by end-stage / fibrosis F4. This is the dominant, robust signal.
- The PERIportal program largely HOLDS its level.
- True DE-ZONATION (gradient flattening with expression retained) is WEAK — and the one view that seemed to show it is partly a z-scoring artifact (explained inside).
- The apparent 'loss of zonation' (anti-correlation collapse) is mostly a DOWNSTREAM consequence of the turn-off: silence the pericentral genes and the coordinate becomes noise.
- The effect TRACKS FIBROSIS specifically, and the label-free ruler is genuine biology.

1. The weak indicators

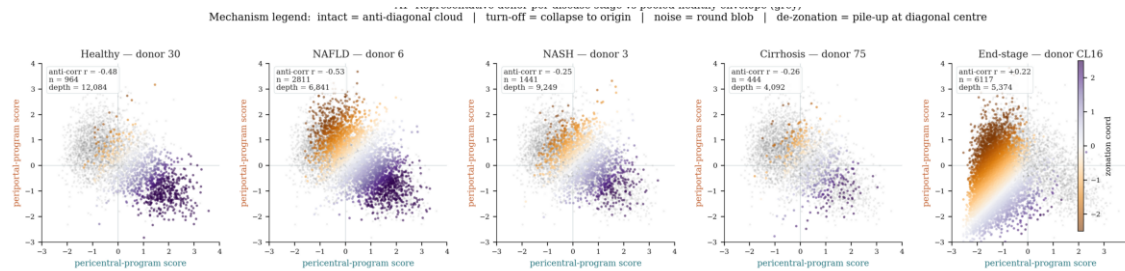
Per representative patient at each stage: the distribution of cells along the zonation axis (top row) and the PC-vs-PP scatter (bottom). This is what 'coordinate spread' actually measures.



Look at the top row: every patient's coordinate distribution is a single broad hump, and its width (SD / IQR, printed under each) changes only modestly and NON-monotonically across stages (NAFLD is actually the widest). A summary like 'spread falls with disease' is an indicator, not a result — which is why we go to the gene-level biology below.

2. The bipolar geometry dissolves

PC-vs-PP scatter for the best-powered representative donor at each stage. In health the two programs trade off (anti-diagonal cloud); with disease that structure is lost.



Healthy (donor 30, $r = -0.48$) and NAFLD/NASH show a tight anti-diagonal — cells are either pericentral-high/periportal-low or vice versa. By Cirrhosis (+0.26) and End-stage (+0.22) the cloud is a round/positive blob; the programs no longer trade off. (This anti-correlation collapse is largely downstream of the pericentral turn-off shown in Part 3.)

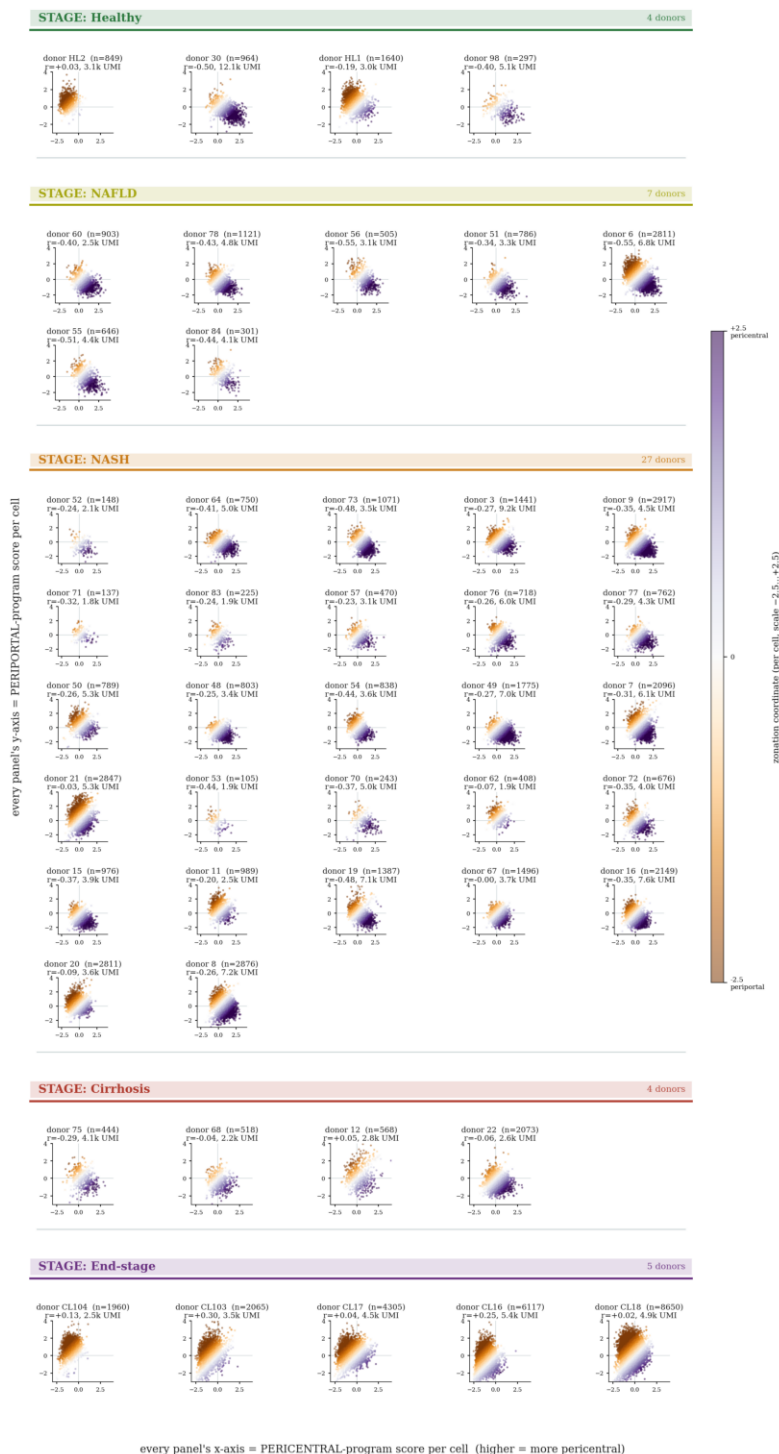
2b. The same, for ALL 47 donors (nothing cherry-picked)

Every donor's PC-vs-PP scatter, grouped into blocks by disease stage. Representatives above are not hand-picked — here is the full cohort.

A1 PC-vs-PP zonation scatter — all 47 donors, grouped by disease stage

Each panel = one donor: x = pericentral-program score, y = periportal-program score, colour = each cell's zonation coordinate (purple = pericentral, orange = periportal). Axes fixed/equal across all donors.

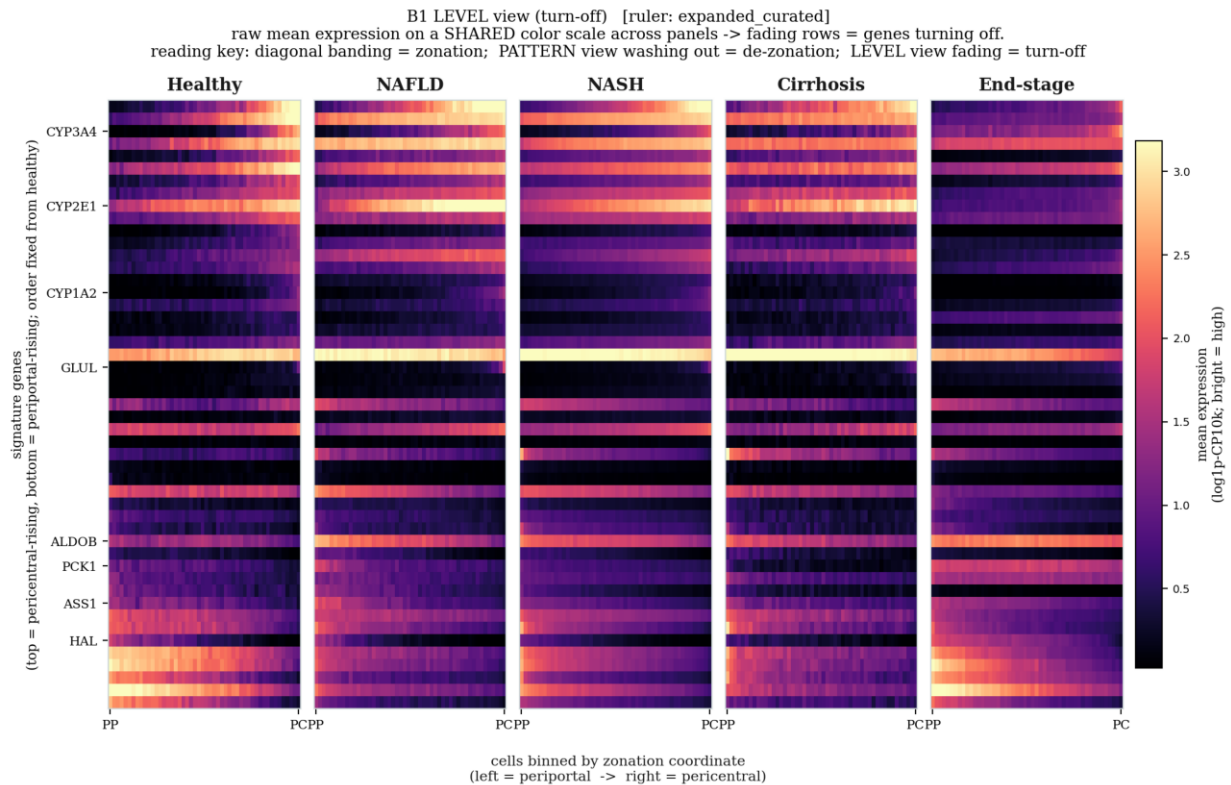
Reading key: anti-diagonal cloud = intact zonation; round / positive blob = collapse. Per panel: n = number of cells in that donor; r = Spearman(pc, pp) anti-correlation; UMI = median SCT-corrected depth.



Read top (Healthy) to bottom (End-stage). The anti-diagonal is clearest in the early blocks and degrades downward, but it is donor-variable — note that two of the four healthy controls are low-depth and already weak. The per-donor anti-correlation r is printed on each panel.

3. The headline: the pericentral program turns OFF

Gene $\times$ zone-ordered-cells heatmap, RAW expression level, shared colour scale across stages (bright = high, dark = low). Genes ordered by their healthy zonal slope (pericentral-rising at top, periportal-rising at bottom).



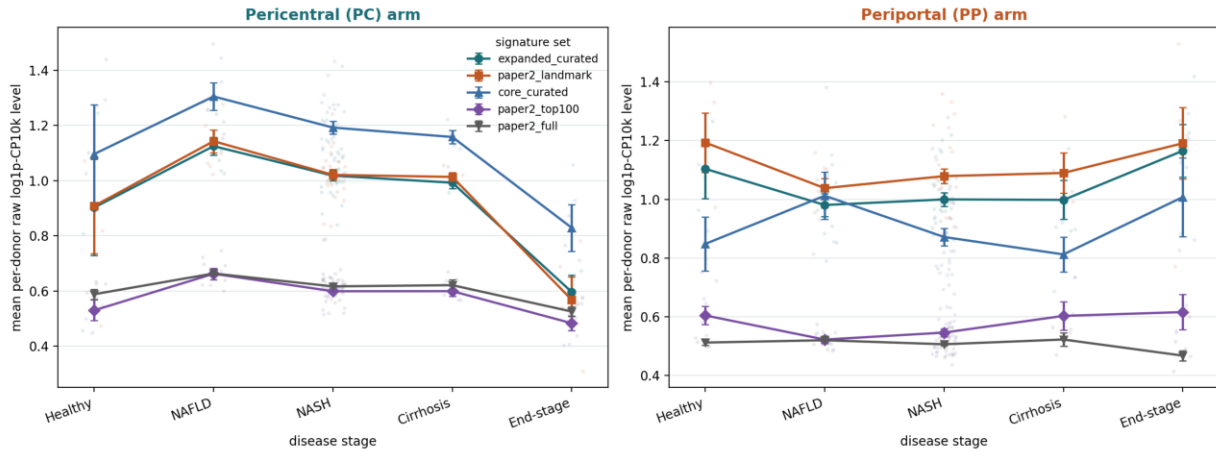
Look at the TOP (pericentral) rows — CYP3A4, CYP2E1, CYP1A2, GLUL: they are bright in Healthy (left) and FADE to dark by End-stage (right). The bottom (periportal) rows stay comparatively bright. This is turn-off, and it is pericentral-biased.

3b. Program level drops — every gene set agrees

Mean per-donor raw expression of the pericentral arm (left) and periportal arm (right) across stages, one line per candidate signature set (a robustness panel).

B4 Program expression LEVEL across disease, by signature set

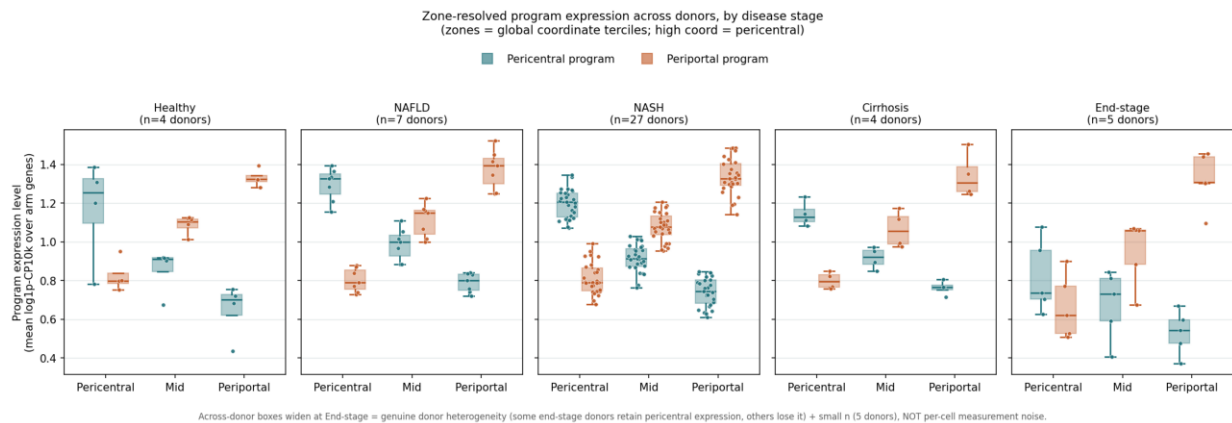
ROBUSTNESS panel: multiple signature sets shown to confirm the direction is set-independent (each line = one candidate PC/PP gene list).
x = disease/fibrosis level; y = mean per-DONOR raw log1p-CP10k expression (un-standardized LEVEL); donor = unit of aggregation; error bars = SEM across donors.
Falling level with disease = TURN-OFF; flat level (with collapsing spatial gradient) = DE-ZONATION.



Left panel: the pericentral lines all DROP at End-stage (−34 to −37%). Right panel: the periportal lines stay flat. The loss is asymmetric and set-independent.

3c. ...and it happens IN the pericentral zone

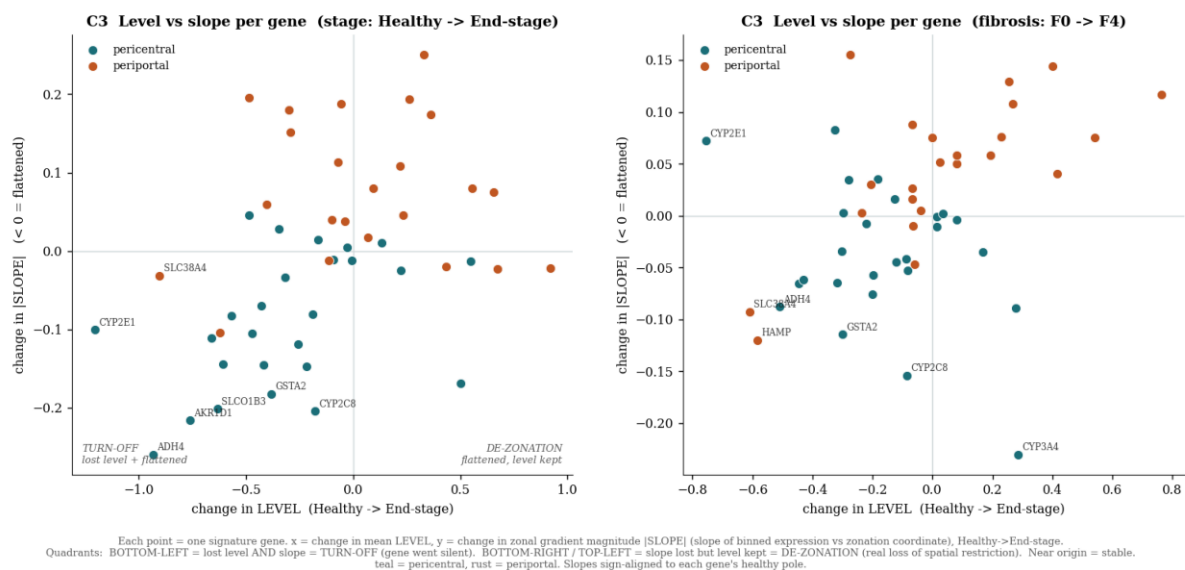
Cells split into three zones (pericentral / mid / periportal by coordinate terciles);
boxplots across donors of each program's level in each zone, per stage.



Teal = pericentral program. In Healthy it is high in the pericentral zone (~1.25) and low in the periportal zone. Follow the pericentral-zone teal box across stages: it collapses to ~0.74 by End-stage. The periportal program (rust) holds its level. (Boxes widen at End-stage = real donor heterogeneity + small n, not measurement noise.)

3d. Gene by gene: turn-off, not de-zonation

For each signature gene, the change in expression LEVEL (x) vs the change in zonal SLOPE magnitude (y), Healthy→End-stage. Bottom-left quadrant = lost both = turn-off.

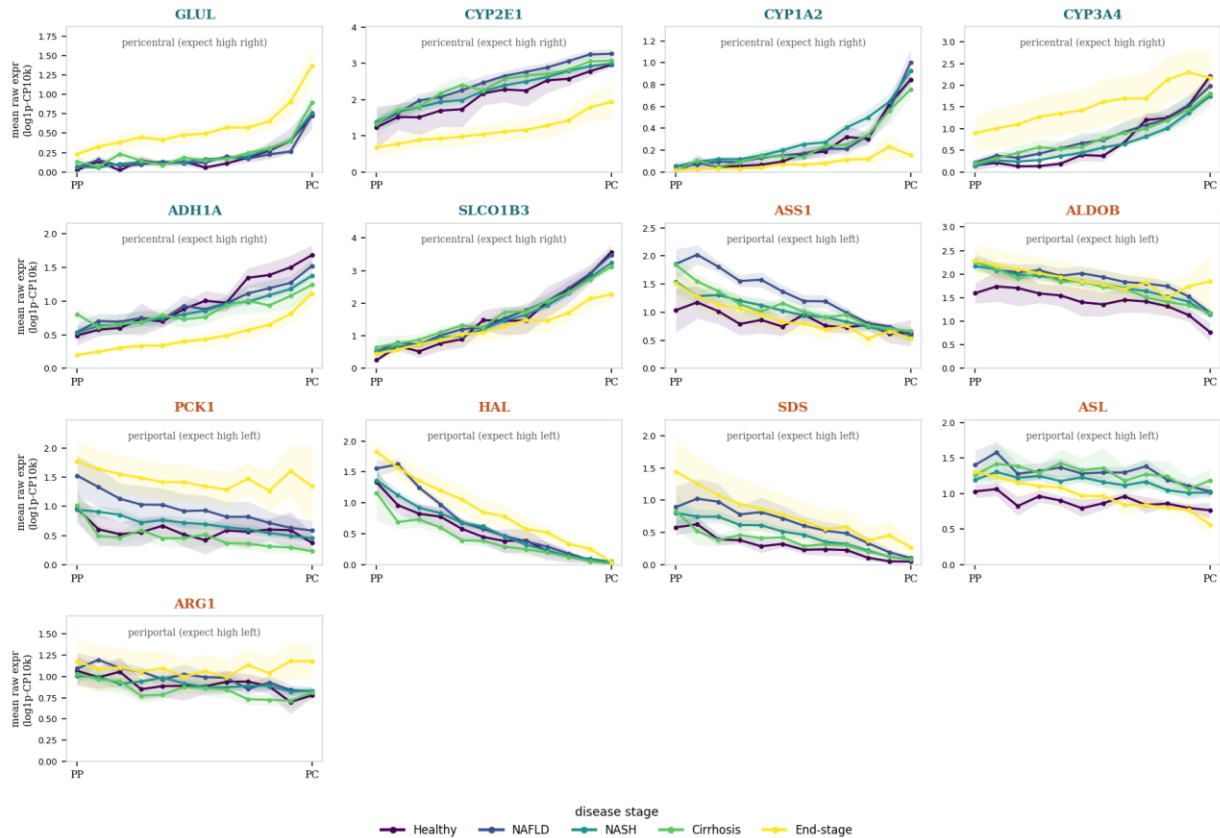


Pericentral genes (teal) cluster in the turn-off quadrant: 14/26 lose BOTH level and gradient (e.g. CYP2E1 dLevel -1.21, ADH4 -0.93). Periportal genes (rust): 22/23 hold their level. The mechanism is turn-off, gene by gene — not a spurious level drop misread as de-zonation.

3e. The marker genes confirm it

Mean raw expression of canonical markers vs the zonation coordinate, one line per stage.

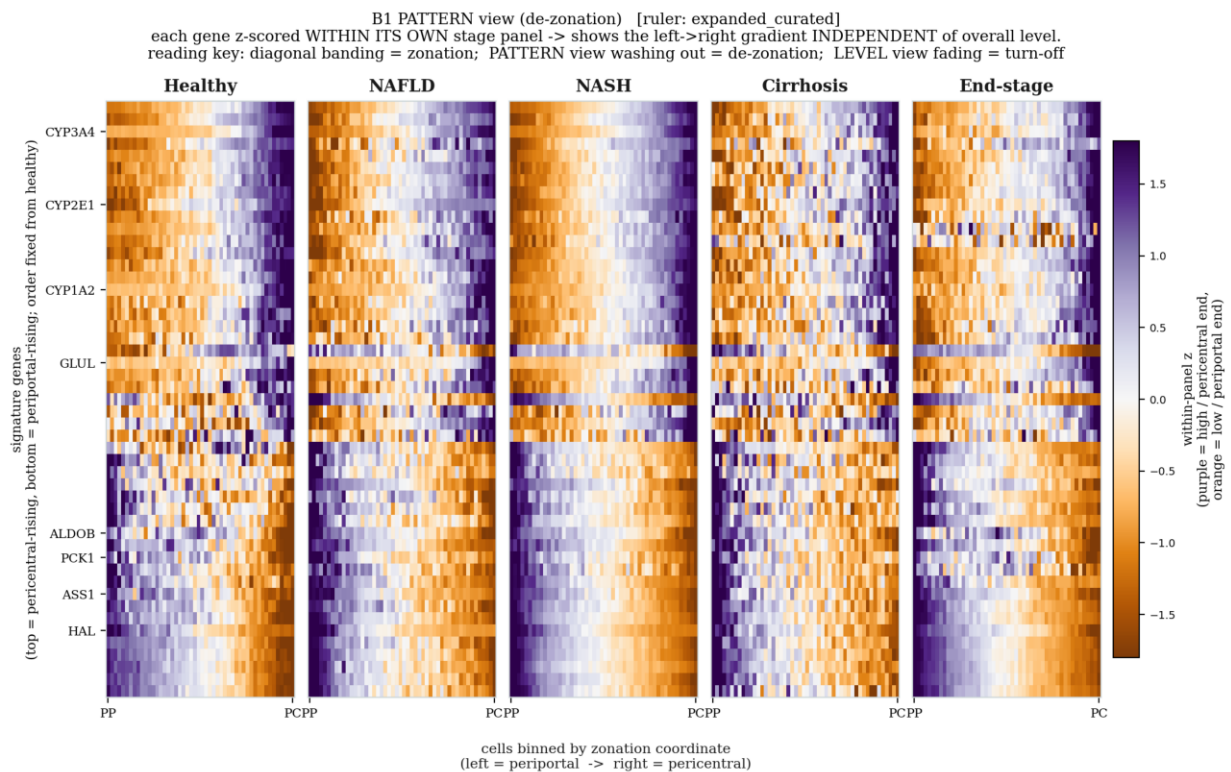
A3 canonical marker spatial profiles across disease [ruler: expanded_curated]
e bin (LEFT=periportal -> RIGHT=pericentral, ~12 donor-averaged quantile bins); y = mean RAW expression (log1p-CP10k). flattening gradient = de-zonation; whole cur



Pericentral markers (CYP1A2, CYP2E1, SLCO1B3, GLUL) keep their gradient through mid-disease then collapse in level at End-stage (turn-off). Periportal markers (ASS1, ALDOB, ARG1) hold their level and mostly flatten only mildly.

4. Is there de-zonation? Honestly, weakly — and here is the tra

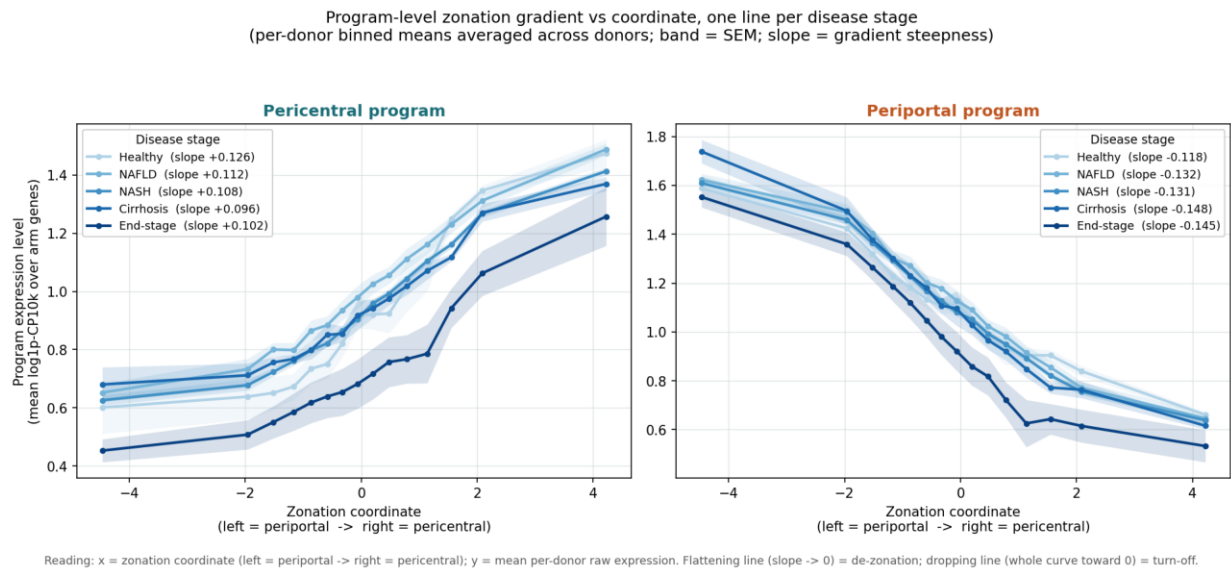
Same heatmap, but each gene z-scored WITHIN its own stage panel (the 'pattern' view, intended to show the gradient independent of level).



The diagonal banding largely PERSISTS through Cirrhosis — so the spatial pattern does not obviously collapse. CRITICAL CAVEAT: in the End-stage panel the top (pericentral) rows show speckled purple/orange. That is NOT a gradient — those genes have turned OFF, so z-scoring their near-zero signal WITHIN the panel amplifies pure noise into fake colour. Read silenced genes from the LEVEL view (Part 3), not here. Net: we do not have strong evidence for de-zonation; the robust event is turn-off.

4b. Program gradients flatten only modestly

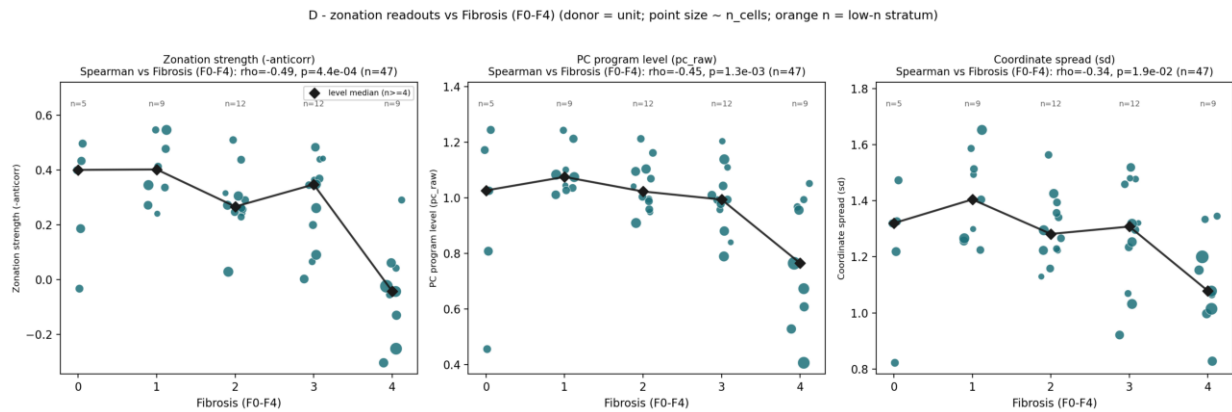
Pericentral (left) and periportal (right) program expression vs the zonation coordinate, one line per stage; the per-stage slope is the steepness of zonation.



The pericentral slope flattens only modestly ($\approx +0.126 \rightarrow +0.102$); the periportal slope barely changes. What moves decisively is the LEVEL (the pericentral curve drops). So de-zonation (pure gradient loss) is the weaker, secondary effect — turn-off dominates.

5. Re-staged by FIBROSIS (not just the coarse 5 stages)

The cohort re-binned by fibrosis grade F0→F4 (available in the metadata). Zonation strength, pericentral program level, and coordinate spread vs fibrosis; per-donor points + per-grade median + Spearman trend.

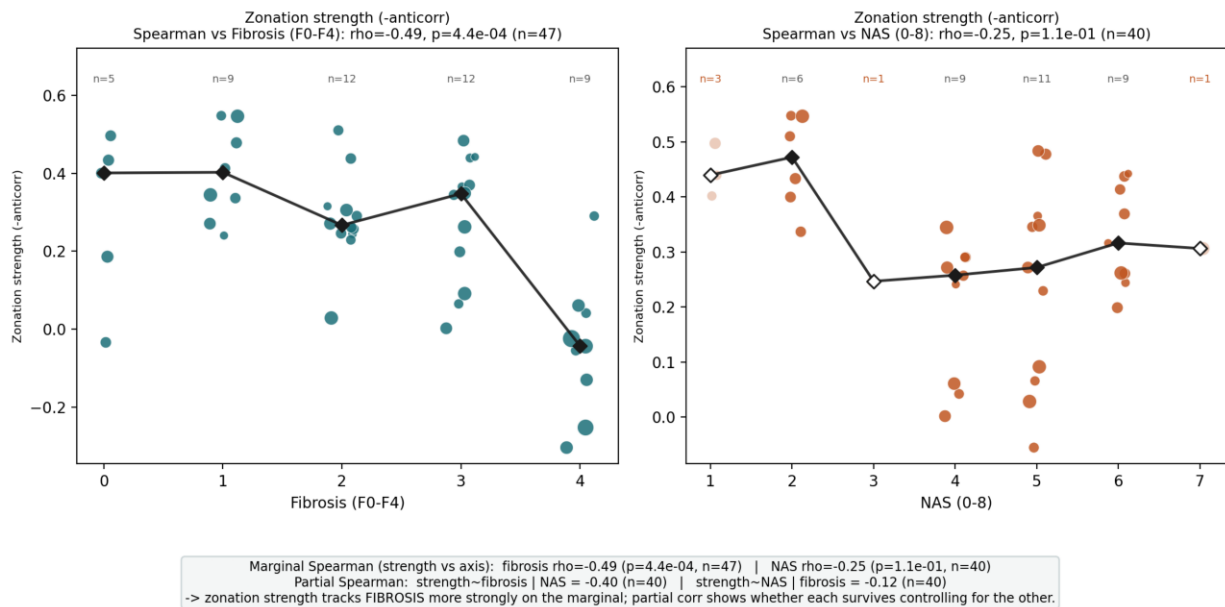


All three decline with fibrosis (strength $\rho=-0.49$, $p=4.4e-4$; pericentral level $\rho=-0.45$). The graded fibrosis axis is a stronger biological readout than the coarse stages. (Low-n grades are faded / flagged on the figure — F0 has 5 donors.)

5b. ...and it is FIBROSIS-specific, not generic activity

Zonation strength vs fibrosis (left) and vs NAS activity score (right), with partial correlations.

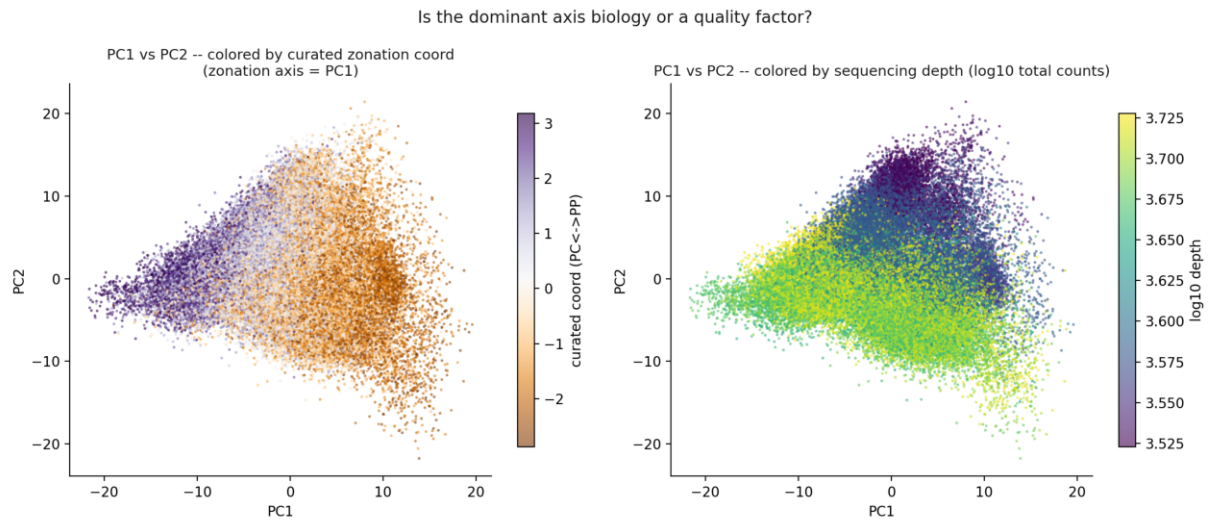
D - which axis does zonation strength track? fibrosis vs NAS



Strength tracks fibrosis ($\rho=-0.49$) far better than NAS ($\rho=-0.25$, n.s.). Partial correlations settle it: controlling for NAS, the fibrosis association holds (-0.40); controlling for fibrosis, the NAS association collapses (-0.12). The signal is tied to FIBROSIS.

6. The label-free ruler is genuine biology

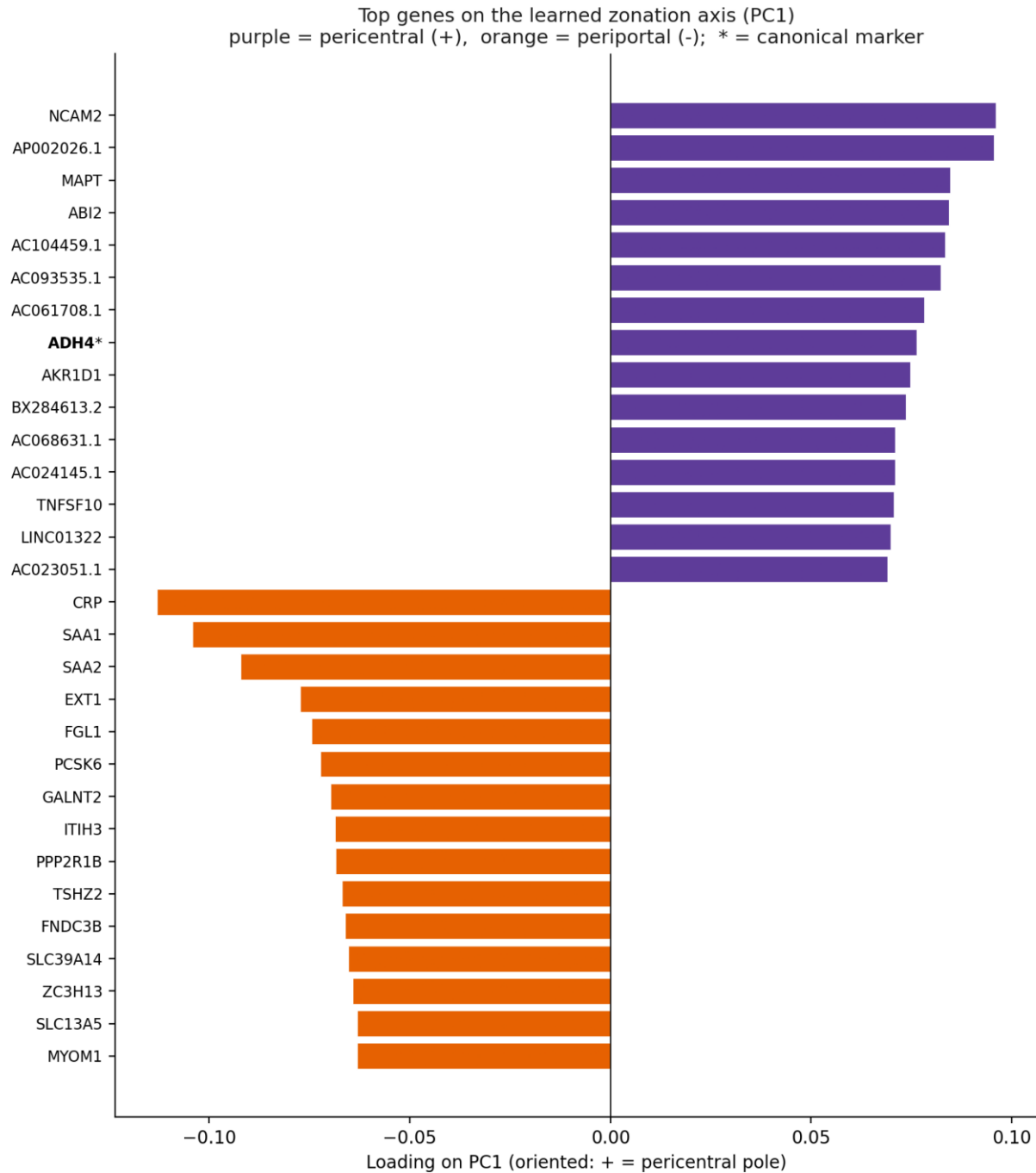
We also built the zonation axis with NO marker genes — PCA on healthy hepatocytes. Cells in PC1-PC2 space, coloured by the curated coordinate (a) and by sequencing depth (b).



PC1 IS zonation: the curated coordinate runs cleanly along it ($|corr|=0.76$; $|corr|=0.54$ with the canonical marker axis; 5.8% of variance, $\sim 2\times$ the next PC). Sequencing depth (panel b) is diffuse and loads on PC2 — NOT on the zonation axis. So the learned ruler is biology, not a technical artifact.

6b. The learned axis's top genes are real zonation genes

Top positive- and negative-loading genes on the zonation principal component.



The periportal pole is dominated by acute-phase/periportal genes (CRP, SAA1, SAA2, fibrinogen-like FGL1); the pericentral pole carries bile-acid/pericentral enzymes (ADH4, AKR1D1). These are textbook zonation markers the PCA found on its own.

Honest caveats

What is strong

Pericentral turn-off is the dominant, robust signal — seen consistently in the level heatmap, the per-set program levels, the zone boxplots, the gene-by-gene level/slope map, and the marker profiles. It tracks fibrosis specifically.

What is weak / unresolved

- DE-ZONATION (gradient flattening with expression retained) is NOT convincingly demonstrated. The program slopes flatten only modestly, and the heatmap 'pattern' view's apparent end-stage signal in pericentral rows is largely z-scoring noise from silenced genes.
- Only 4 healthy-control donors, two of them low-depth and weakly zoned — the healthy baseline is thin.
- The scored matrix is SCT-corrected (depth pre-equalized), so this is not built on raw-depth-confounded data; a true raw-depth check would need the RNA assay.
- Confounder controls are intentionally PARKED (they had been validating the secondary indicators, not the turn-off) — see CONFOUNDERS_PLAN.md for the recentered plan and stronger checks (negative-control genes, purity, covariate regression).
- This is a cross-sectional human cohort: we establish a robust ASSOCIATION between disease/fibrosis and pericentral turn-off, not proof that disease causes it.